12. Serial Communications

Data can be transferred between one or more analyzers, and a single host device, using the Modbus Remote Terminal Unit (RTU) protocol. The host device sends polls, to which the analyzer replies.

12.1. Serial Communications Specifications

Specification	Description	
Supported commands	Command 3	"Read Output (Holding) Registers"
	Command 6	"Preset Single Register"
	Command 16	"Preset Multiple Registers."
Command 16 limit	Limited to one register in length (special case).	
Command 3 limit	Limited to 127 registers in length.	
Broadcast mode	Supported for commands 6 and 16.	
Serial port hardware	Half-duplex RS485	
Data format	8 bits, no parity, 1 stop bit	
Baud rates	Baud rates from 300 to 19,200 are supported	
Standard baud rate	9600	
Maximum poll rate	Eight per second at 9600 Baud, sixteen per second at 19200 Baud	
Receive-transmit delay	Minimum: 5 characters between receipt of poll and reply.	
	Maximum: 250 milliseconds, if maximum poll rate is reached	
Error handling	Invalid start register address	Command ignored, no reply
	Read more than 127 registers	Command ignored, no reply
	CRC error, poll corrupted	Command ignored, no reply
	Read over page boundary	Command ignored, no reply
	Read invalid page	Command ignored, no reply
	Data written outside valid limits	Reply is made but data is ignored
	Read of unsupported register	00H Data returned in reply

12.2. Serial Communications Notes

The modbus master should be secured from unauthorized or unintentional changes. Networked access is not recommended.

WARNING! Serial communications must not be used for critical safety functions such as process shutdown.

Writing illegal data outside the register's acceptable limits is acknowledged with a reply, but the data is ignored. The modbus master should verify completion of write commands by a subsequent read of the register.

Data written to the COMMANDS register is cleared as soon as the command is executed. Reading the COMMANDS register most likely returns 00H data, even if the register has just been written. The host should verify COMMANDS by reading the STATUS register.

Data written to non-volatile registers are saved to the EEPROM sixty seconds later. When several write commands are received consecutively, that is, with less than 60 seconds between them, the EEPROM is written 60 seconds after the last command. The SAVE ALL command saves to the EEPROM immediately.

Some modbus master devices prefer to receive a reply with the register data formatted with the low order byte transmitted first. Read registers at pages 4 or 5 to get register data formatted –lo:hi."

Page#	Hexadecimal addresses	Accessible contents
00	0000H through 00xxH	Registers 1 through 64
01	0100H through 01xxH	Registers 257 through 272
04	0400H through 04xxH	Copy of registers 257 through 272, lo:hi format, effective register numbers 1025 through 1040
05	0500H through 05xxH	Copy of registers 1 through 64, lo:hi format, effective register numbers 1281 through 1344

12.3. Register Conversion Factors

After it is acquired, the serial port data is converted into engineering units using a conversion factor. For the following registers, the data is a signed 16-bit word. Negative numbers are in two's complement form. To convert the register data, multiply by the conversion factor in the table.

REGISTERS	Format	Conversion Factor
READING PEAK HI PEAK LO CAL RDNG LO ALARM HI ALARM MA TEST MA CAL MA FAULT 4MA RDNG 20MA RDNG	Reading	Multiply by 0.0390625
MA TEST MA CAL MA FAULT	Reading format expressed as milliamps	Multiply by 0.000625, then add + 4.0 Milliamps
HSG TEMP SET TEMP FLAME RAW ZERO RAW SPAN SPAN °C	Temperature	Multiply by 0.24426 °C
CELL VAC SET VAC	Cell Vacuum	Multiply by 0.351734
4 MA ADJ 20 MA ADJ	Milliamps	0.1 Milliamps
ASP PRES	Pressure	Multiply by 0.105862 PSIG
VARIOUS	ON/off	ON = 1, active; off = 0, inactive

12.4. Registers

	Register Name	#	Register Contents *Registers that can be written are indicated by this symbol: ✎.*
GAUGES MENU	READING	1 *50*	The reading Register *50* is a copy of READING that includes the effect of MA CAL, MA FAULT and MA TEST register functions.
	STATUS	2	Device status messages, codes, and their meanings **Message / Code / Meaning** Normal / 0 / Normal operation, no alarms Lo Alarm / 1 / Low alarm, reading below level Hi Alarm / 2 / High alarm, reading above level Downscale / 3 / Fault, reading lower than –10%LFL Zero On / 4 / Zero calibration or zero test in progress Span On / 5 / Span calibration or span test in progress Purging / 6 / Purging calibration gases Cal ends / 7 / Calibration ending Zero fail / 8 / Zero calibration failed, result unacceptable Span fail, low / 9 / Span calibration failed, result too low Span fail, high / 10 / Span calibration failed, result too high Warm-up / 11 / Power-on time delay for self diagnostics RAM test / 12 / Self diagnostic of RAM RAM fail / 13 / RAM has failed self diagnostic test Data lost / 14 / Checksum for nonvolatile data in EEPROM failed EE fail / 15 / Self diagnostic check of EEPROM failed Vac low / 16 / Sample vacuum too low Vac high / 17 / Sample vacuum too high Temp Low / 18 / Temperature of analyzer is too low. Temp Hi / 19 / Temperature of analyzer is too high. A/D fail / 24 / Self check of A/D converter failed. TC2 Fail / 25 / Thermocouple for temperature control failed. Flameout / 26 / Sensing flame is out. ROM fail / 27 / Continuous self-diagnostic of ROM has failed.
	PEAK HI	38	Highest reading taken since last reset command
	PEAK LOW	39	Lowest reading taken since last reset command
	HSG TEMP	9	Actual flame cell housing temperature in Degrees Centigrade
	SET TEMP ✎	24	Flame cell housing temperature set point in Degrees Centigrade SNR692: Min 100°C, Max 153°C, Default 120°C
	CELL VAC	10	Sample vacuum in inches of water
	SET VAC ✎	23	Setting for control of sample vacuum, typically 40 inches of water..
		42	Do not write to this register via modbus.
	FLAME	11	Flame temperature (Thermocouple TC1 reading) in Degrees Centigrade
	ASP PRES	33	E/P controller, PSIG required to obtain sample vacuum setting

	REGISTER NAME	#	REGISTER CONTENTS
CAL MENU	RAW ZERO	29	Flame temperature from zero calibration in degrees Centigrade Minimum value is 200°C Maximum value is 300°C Default value is 250°C.
	RAW SPAN	30	Flame temperature from span cal minus the raw zero, in degrees Centigrade.
	CAL TIME ✎	28	Duration of each calibration interval in minutes Minimum setting 1 minute, maximum 4 minutes. Default is 1 minute.
	CAL RDNG ✎	7 *43*	Reading for span calibration. Minimum value is ___ Maximum value is ___ Register 43 is a copy of CAL RDNG from the last successful calibration.
	SPAN °C ✎	56	Setting for expected RAW SPAN temperature rise during span calibration
	SPEED ✎	20	Speed of response and noise filter Off 0 Slowest setting. Low 1 Med 2 High 3 Fastest setting
	CAL LOCK	60	Indication of calibration lock. Read only. Hardware Lock Bit 0 Software Lock Bit 1
COMMANDS MENU	COMMANDS ✎	8	Execute a command to control the device - 0 (no commands, inactive) Reset 200H Reset alarms Acknowledge 100H Silence the horn, clear “service needed” relay
			Calibrate analyzer with gas, adjust settings, activate faults if response unacceptable Zero Cal 300H Perform zero calibration and adjust zero Span Cal 400H Perform span calibration and adjust span Full Cal 500H Perform zero and span calibration and adjust
			Test analyzer response with gases, but do not adjust the settings, ignore faults Zero Test 600H Test zero with gas, do not adjust Span Test 700H Test span with gas, do not adjust Full Test 800H Test zero and span, do not adjust
			Commands affecting EEPROM Save All 900H Save all settings to EEPROM Cold Start 0A00 Force device into COLD START condition. Do not use.
			Commands for remote control - not on menu 0B00H Emulates “Menu” pushbutton 0C00H Emulates “Select” pushbutton 0D00H Emulates “Flashlight” activation
			Commands for security - not on menu - 0E00H “Momentary lock” - internal use only. Do not use. Lock 0F00H Locks out calibration, software lock. Unlock 1000H Unlocks calibration.

	REGISTER NAME	#	REGISTER CONTENTS
OUTPUTS MENU	RLY TEST	45	Relays activated for test. Format is identical to RELAYS register. Test is run through menu access only - do not write to this register with modbus (serial communication).
	MA TEST	44	Circuit test of milliamp output, in milliamps, increases output only, deactivates upon leaving menu. Test is run through menu access only - do not write to this register with modbus (serial communication).
	4 MA RDNG	13	Reading for which 4 milliamps is output.
	20 MA RDNG	14	Reading for which 20 milliamps is output.
	4 MA ADJ	25	Circuit adjustment of milliamp output at the 0%LFL (4 milliamp) point – factory set Range is +/- 2 milliamps. Default value is 0.
	20 MA ADJ	31	Circuit adjustment of milliamp output at the full scale (20 milliamp) point – factory set. Range is +/- 2 milliamps. Default value is 0.
	MA CAL	26	Value for milliamp output during calibrations Minimum is 2.5 milliamps, maximum 22.7 milliamps, default is 22.1 milliamps.
	MA FAULT	32	Value for milliamp output during faults Minimum is 2.5 milliamps, maximum 22.7 milliamps, default is 21.0 milliamps.
	CONTROL1	36	Assigns a COMMAND to the CONTROL1 input. (See COMMANDS register)
	CONTROL2	37	Assigns a COMMAND to the CONTROL2 input. (See COMMANDS register)
	RLY CNFG	58	Configuration setting for the HORN relay 00 Default setting. The relay is used for the HORN function. 01 AIR/FUEL function. The relay energizes (contacts close) when flame cell is at proper operating temperature. 02 OVERRIDE function. Relay is always energized (contacts always closed)
COM PORT MENU	SET BAUD	22	Default Baud Rate Setting (See below for allowable settings and Baud rates)
	BAUD RATE	47	Baud Rate Currently In Use. Can differ from the default setting if AUTOBAUD is "on." The allowable settings are: 0010H 19200 0011H 9600 Standard value is 9600 Baud 0012H 4800 0013H 2400 0014H 1200 0015H 600 0016H 300
	AUTOBAUD	48	Control for automatic baud rate detection feature (vs. fixed baud rate setting) 0 Off Baud rate fixed, equals "SET BAUD" value 1 On Device automatically seeks proper baud rate
	ADDRESS	21	Port address of device Valid settings are 1 to 255, Default setting is 1. Note: the address in the modbus data packet is one less than that stored in the address register "address 1 = 0000H"
	ID NBR	19	User-assigned identification number. Valid settings are 0 to 99. Default setting is 0.

	REGISTER NAME	#	REGISTER CONTENTS
ALARMS MENU	LO ALARM ✎	5	The alarm level, in the same format as reading Default setting is _____ Adjustable from ____ to ____
	HI ALARM ✎	6	The alarm level, in the same format as reading Default setting is _____ Adjustable from ____ to ____.
	LATCHES ✎	17	Alarms that will latch until manually reset, versus those that will auto reset Format is identical to RELAYS register.
	FAILSAFE ✎	18	Relays that are failsafe (normally energized) and which de-energize upon activation of the alarm. FAULT is always failsafe. Format is identical to RELAYS register. Do not write to this register via modbus - change only through the operator panel.
	AUTO IGN ✎	35	Automatic Re-ignition setting 0 = None, single attempt to ignite is made upon flameout, after reset. 1 to 120 = Number of re-ignition attempts at 15 second intervals. Default setting is 0, disabled.
	VERSION		Software version (for complete Version header see below registers 257 - 272)
	TYPE	62	Analyzer type - This register is written once at factory
	SERIAL	63	Serial number – displayed as year of production and serial, eg, 02-101 for year 2002 serial 101. Data in this register is the serial only. The year of production is stored in the ASCII text string registers 257 to 272. This register is written once, at the factory upon shipment.
	LANGUAGE ✎	64	Display language 0 = English (default) 1 = German

REGISTER NAME	#	Register Contents **Note:** These registers are accessible through the serial port In some prior software versions LINEAR and MA SCALE were on menus.
FULLSCLE	3	Full scale measurement range. 1000 for standard units.
RELAYS	4	Logical state of the control relays. This data matches the corresponding LED indicators. A bit of 1 = “active”, 0 = “inactive” Data is in binary format. Packed flags: 0000 0000 TISZ HFDW T Heater I Ignition S Span solenoid valve Z Zero solenoid valve H Horn relay F Fault relay D Hi Alarm relay W Low Alarm relay
	49	Copy of RELAYS in low byte position and FAILSAFE register in high byte position.
DUMMY	46	A register that performs no function, but can be used to debug serial communications
A/D REF	16	The reference voltage of the A/D converter. Should be 506 ± 25.
SERVICE	59	SERVICE NEEDED indication. A bit of 1 = “active”, 0 = “inactive” Data is in binary format. Packed flags: 0000 0000 A0RD LSZF F N/A Z CHK_ZERO S CHK_SPAN L CHK_LED - controls flashing of fault LED when any CHK_ bit is set D CHK_DATA R CHK_RDNG A AIR_FUEL
VERSION	257 thru 272	Device identification, in ASCII text string. „ BTU „ Analyzer Model „VER 5.xx“ Software version „SNR69x „ Analyzer assembly number “ 2007 „ Analyzer serial number – year of production
	40	Internal pointer used for display of version - do not use.
DISPLAY	51 thru 55	Copy of the LCD display buffer in ASCII.